

FOSSEE

eSim Semester Long Internship — Autumn 2026

PCB DESIGN PROPOSAL

Candidate Name	Sinjini Biswas
Project	SpectraSense
Design	Two-layer LM741-based signal-conditioning PCB

1. Design Idea

SpectraSense is proposed as a compact, two-layer, through-hole analog PCB centered on an LM741 operational amplifier. The design provides a dedicated sensor-input interface, an analog signal-conditioning stage with resistive feedback, an auxiliary diode branch, and a separate power-entry interface with local decoupling.

The PCB is intended as a practical prototype-style implementation of the corresponding eSim circuit. Standard through-hole components are used so that the board remains straightforward to inspect, assemble and test.

2. Basic Functionality

The sensor signal enters through J1 (SENSOR_IN) and is applied to the LM741 inverting-input/summing node. R1, with a value of 100 kΩ, provides the feedback path from the op-amp output back to this node, forming the main inverting signal-conditioning stage. The non-inverting input is referenced to ground.

J2 (POWER_IN) provides the board power-entry interface. C1 (100 nF) is connected across the power and ground nodes to provide local bypass decoupling. D1 (1N4148) is implemented as a separate auxiliary branch. The schematic also contains simulation-only sources used for ngspice analysis; these are not physical PCB components.

3. Functional Architecture

Block	Function	Physical Implementation
Sensor Input	Accepts the external sensor signal and ground reference.	J1, 2-pin THT header
Signal Conditioning	Inverting LM741 stage with resistive feedback.	X1 LM741 DIP-8 + R1 100 kΩ
Auxiliary Branch	Separate diode branch represented in the schematic.	D1 1N4148, DO-35 THT
Power / Decoupling	Provides power entry and local bypassing.	J2 + C1 100 nF

4. PCB Implementation Approach

The PCB is organized according to the functional structure of the circuit. The external connectors are kept toward the board edges for accessibility, the LM741 is placed centrally, and R1 is positioned close to the amplifier feedback region. C1 is kept close to the power-entry area so that the decoupling connection remains short.

The design uses both F.Cu and B.Cu to satisfy the two-layer requirement and to keep the connections direct and readable. The board outline is approximately 45.72 mm × 25.40 mm (1.8 in × 1.0 in).

5. Component and Footprint Strategy

Reference	Component	Value	Package
X1	LM741	Op-amp	DIP-8 THT
R1	Resistor	100 kΩ	Axial THT
D1	Diode	1N4148	DO-35 THT
C1	Capacitor	100 nF	Disc THT
J1	Pin Header	SENSOR_IN	1×2, 2.54 mm THT
J2	Pin Header	POWER_IN	1×2, 2.54 mm THT

6. Verification Plan

The completed design is intended to be verified through the standard eSim PCB workflow: schematic-to-PCB transfer, footprint and placement inspection, two-layer routing, Design Rule Check (DRC), and 3D Viewer inspection.

The final implementation provides evidence for these stages through the schematic capture, footprint mapping, routed PCB layout, DRC result and 3D visualization included in the accompanying Task 7 report.

7. Expected Outcome

The proposed outcome is a compact, organized and manufacturable two-layer through-hole PCB that physically implements the SpectraSense circuit. The layout is intended to maintain accessible external connections, short local connections around the signal-conditioning stage and power decoupling, while remaining compatible with the required eSim PCB workflow.

This proposal describes the PCB design idea and its basic functionality; the detailed implementation evidence and verification results are provided separately in the Task 7 PDF report.